

INTEGRATIVE PHYSIOLOGY (POST LEC)

Lecture #2: CIRCULATORY & MUSCLE | OPHT 1-1 | MIDTERMS

By Accel Jimenez

PHYSICAL CHARACTERISTICS OF CIRCULATION

Systemic Circulation - aka greater circulation or peripheral circulation.

Pulmonary Circulation - supplies the lungs

Arteries

- Transport blood under high pressure to the tissues
 - the arteries have strong vascular walls, and blood flows at a high velocity in the arteries

Arterioles

- Last small branches of the arterial system
- Act as control conduits through which blood is released into the capillaries
- Altering blood flow (constrict or dilate depending on the need)

Capillaries

- Exchange fluid, nutrients, electrolytes, hormones, and other substances between the blood and the interstitial fluid
- Capillary walls are thin
- Capillary pores permeable to water and other small molecular substances

Venules

- Collect blood from the capillaries and gradually coalesce into progressively larger veins

Veins

- Conduits for transport of blood from the venules back to the heart
- Major reservoir of extra blood
 - Thin walls
 - Low pressure

VASCULAR DISTENSIBILITY

Vascular Distensibility (definition & formula):

- Expressed as the **fractional increase in volume per mmHg rise in pressure**.
- Formula:
- $\text{Vascular distensibility} = \frac{(\text{increase in volume})}{(\text{increase in pressure} \times \text{original volume})}$
- Example: If a vessel with 10 ml blood increases by 1 ml at +1 mmHg → distensibility = **0.1 per mmHg (10% per mmHg)**.
- The **formula quantifies vascular compliance** relative to pressure changes.

All blood vessels are distensible (expandable)

- This property allows them to change volume with changes in pressure.

Arteries:

- Their distensibility lets them **accommodate the pulsatile output of the heart**.
- They **average out pressure pulsations**, ensuring smooth, **continuous blood flow** in small vessels.

Veins (most distensible)

- Even a **slight increase in venous pressure** allows the vein to **store 0.5-1.0L of extra blood**.
- Veins act as a **blood reservoir**, releasing stored blood when needed by circulation

Veins vs Arteries

- Veins are **~8× more distensible than arteries**.
 - **Thinner, weaker walls** → larger volume change with same pressure increase.
- Pulmonary circulation:
 - **Pulmonary veins** → similar distensibility to systemic veins.
 - **Pulmonary arteries** → operate at ~1/6 systemic pressure → **~6× more distensible** than systemic arteries.

VASCULAR COMPLIANCE

- Definition: Total volume change per mmHg pressure rise.
- $\text{Compliance} = \Delta \text{"Volume"} / \Delta \text{"Pressure"}$
- Key distinction:
 - **Distensibility** = % volume change relative to original volume
 - **Compliance** = absolute volume change per pressure change
- A vessel can be highly distensible but still have low compliance if its volume is small.
- **Systemic veins:**
 - **Compliance ≈ 24× greater than arteries** (8× more distensible × 3× larger volume).

VOLUME-PRESSURE CURVES

- **Arterial System**
 - ~700 mL blood → mean arterial pressure ≈ 100 mmHg.
 - If volume drops to ~400 mL → pressure falls to zero.

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- Venous system:
 - Normal volume: **2000–3500 mL**.
 - Several hundred mL volume change → venous pressure shifts only 3–5 mmHg.
 - Explains why up to 0.5 L transfusion causes minimal circulatory disturbance.

EFFECT OF SYMPATHETIC STIMULATION/INHIBITION

- Sympathetic stimulation:
 - ↑ Vascular smooth muscle tone → **higher pressure at same volume** (arteries & veins).
 - Allows blood to shift from veins → heart → ↑ cardiac output.
 - Critical during **hemorrhage**: maintains circulation even with **up to 25% blood volume loss**.
- Sympathetic inhibition:
 - ↓ Tone → lower pressure at each volume.

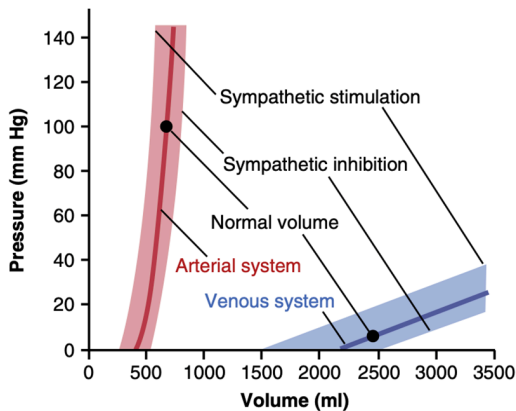


Figure 15-1. Volume-pressure curves of the systemic arterial and venous systems, showing the effects of stimulation or inhibition of the sympathetic nerves to the circulatory system.

DELAYED COMPLIANCE (STRESS-RELAXATION)

- When volume suddenly ↑:
 - Immediate elastic distention → ↑ pressure.
 - Over minutes–hours: **smooth muscle slowly relaxes**, pressure falls toward normal.
- When volume ↓:
 - Vessels gradually contract, helping stabilize circulation.
- Physiological importance:
 - Allows adaptation to extra blood (e.g., transfusion).

- Helps adjust circulation to blood loss (hemorrhage).

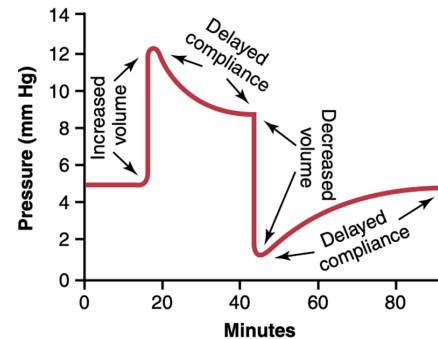


Figure 15-2. Effect on the intravascular pressure of injecting a volume of blood into a venous segment and later removing the excess blood, demonstrating the principle of *delayed compliance*.

Key Takeaways:

- **Veins = primary reservoir** → highly distensible + high compliance.
- **Compliance = distensibility × volume** (not interchangeable).
- **Sympathetic tone** regulates blood distribution & maintains function during stress (e.g., hemorrhage).
- **Delayed compliance** is an adaptive mechanism for both excess and reduced blood volume.

Feature	Arteries	Veins
Wall structure	Thick, strong, muscular	Thin, weaker
Distensibility	Low (resist expansion)	High (~8× more than arteries)
Compliance ($\Delta V/\Delta P$)	Low	Very high (~24× more than arteries)
Volume capacity	Smaller (~700 ml in systemic arteries)	Large (~2000–3500 ml in systemic veins)
Role in circulation	Maintain pressure & smooth out heart pulsations	Reservoir for blood storage
Response to pressure changes	Small volume change for given ΔP	Large volume change for small ΔP
Sympathetic stimulation	↑ Tone → ↑ arterial pressure	↑ Tone → shifts blood to heart , supports circulation in hemorrhage
Pulmonary system	Pulmonary arteries: ~6× more distensible than systemic arteries	Pulmonary veins: similar to systemic veins
Adaptation (Delayed Compliance)	Limited effect	Strong stress-relaxation response → adjusts pressure after transfusion or hemorrhage

ARTERIAL COMPLIANCE & CONTINUOUS FLOW

- With each heartbeat, arteries fill with blood.
- **Arterial distensibility/compliance smooths pressure pulsations** → prevents blood from flowing only during systole.

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- By the time blood reaches capillaries → **flow is continuous with minimal pulsation.**

Normal Pressure Values

- **Systolic pressure (SP):** ~120 mmHg.
- **Diastolic pressure (DP):** ~80 mmHg.
- **Pulse pressure (PP):** $SP - DP = \sim 40$ mmHg.

Determinants of Pulse Pressure

- Two major factors:
 - **Stroke volume output** ($\uparrow$ SV → $\uparrow$ PP).
 - **Arterial compliance** ($\downarrow$ compliance → $\uparrow$ PP).
- Equation:
- "Pulse pressure" = "Stroke volume" / "Arterial compliance"
- **Age effect:** Arteries stiffen with arteriosclerosis → **PP may double in old age.**

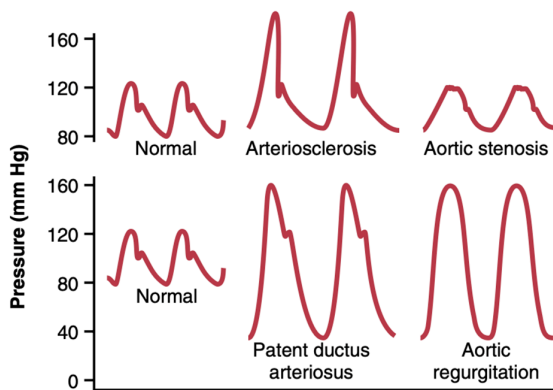


Figure 15-4. Aortic pressure pulse contours in arteriosclerosis, aortic stenosis, patent ductus arteriosus, and aortic regurgitation.

Abnormal Pulse Pressure Contours

- **Aortic stenosis:** Narrow valve → $\downarrow$ stroke volume → $\downarrow$ PP.
- **Patent ductus arteriosus:** Blood leaks from aorta → pulmonary artery → low diastolic pressure + widened PP.
- **Aortic regurgitation:** Blood flows back into LV → aortic pressure may fall to zero between beats, no diastolic notch (incisura).

Transmission of Pressure Pulses

- Pressure wave travels along arteries as a moving wave, not actual bulk blood movement.
- Velocity of transmission:
 - Aorta: 3–5 m/s (slow due to high compliance).

- Large branches: 7–10 m/s.
- Small arteries: 15–35 m/s (fast due to low compliance).

- Transmission velocity = **15× faster than blood flow velocity.**

Damping of Pressure Pulses

- Pulsations **diminish in smaller arteries, arterioles, and capillaries.**
- Damping factors:
 - **Resistance** → **limits forward flow.**
 - **Compliance** → **requires more blood to raise pressure.**
- **Damping = Resistance × Compliance.**
- Result: Capillaries usually experience **steady flow, not pulsatile.**

Mean Arterial Pressure (MAP)

- Definition: The average arterial pressure **measured over time** (millisecond by millisecond).
- Not simply the average of systolic (SP) and diastolic pressure (DP).
- Why $MAP \neq (SP + DP)/2$
 - The heart spends more time in diastole than systole at normal heart rates.
 - Therefore, **MAP is closer to diastolic pressure.**

Weighting of Pressures

- At normal heart rates:
 - **60% influenced by diastolic pressure**
 - **40% influenced by systolic pressure**
- Formula approximation:
 - $MAP \approx DP + 1/3 (SP - DP)$
- **Effect of Heart Rate**
 - **Normal rate:** MAP closer to DP.
 - **Very high heart rate:** Less time in diastole → **MAP shifts closer to the average of SP and DP.**

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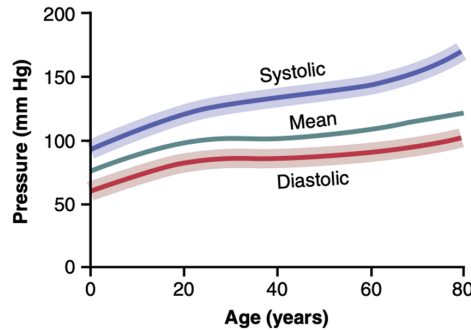


Figure 15-8. Changes in systolic, diastolic, and mean arterial pressures with age. The shaded areas show the approximate normal ranges.

- Example: Blood Pressure = 120/80 mmHg
- **SP = 120 mmHg**
- **DP = 80 mmHg**
- **PP = 120 – 80 = 40 mmHg**
- **Now substitute into the formula:**
 - **"MAP" = $80 + 1/3 (40)$**
 - **"MAP" = $80 + 13.3 \approx 93$ mmHg"**
- This shows MAP is closer to diastolic (80) than systolic (120), since the heart spends more time in diastole.

Key Takeaways:

- Pulse pressure reflects stroke volume & arterial compliance.
- Veins = **volume reservoir**; arteries = **pressure regulators**.
- Arterial compliance ensures steady capillary flow.
- Abnormal pulse contours (stenosis, PDA, regurgitation) provide diagnostic clues.
- Damping ensures **tissues receive continuous blood supply**.
- MAP is the **best indicator of tissue perfusion pressure**.
 - It is weighted toward diastolic pressure at normal rates but approaches the average of SP and DP at high heart rates.

Functions of Veins

- Passageways for blood return to the heart.
- Can constrict or dilate → serve as blood reservoirs (store/release blood when needed).
- Contain the venous pump → **propels blood forward**.
- Help regulate cardiac output by influencing venous return.

Central Venous Pressure (CVP) = Right Atrial Pressure (RAP)

- **Normal:** ~0 mmHg (equal to atmospheric)
- **Determined by balance between:**
 - Heart's pumping ability (↓ RAP if strong; ↑ RAP if weak).
 - Venous return (↑ RAP with more inflow).
- Factors that ↑ venous return (and RAP):
 - Increased blood volume.
 - Increased venous tone → ↑ peripheral venous pressure.
 - Arteriolar dilation → rapid flow into veins.
- **Abnormal ↑ RAP:** 20–30 mmHg in heart failure or massive transfusion.
- Lower limit: –3 to –5 mmHg (with strong pumping or severe hemorrhage).

VENOUS RESISTANCE & PERIPHERAL VENOUS PRESSURE

- **Large veins:** very low resistance when distended.
- But **compression points** (ribs, organs, neck, abdomen) increase resistance.
- **Peripheral venous pressure (lying down):** +4 to +6 mmHg higher than RAP.
- **High RAP (> +4–6 mmHg)** → backup of blood, vein enlargement, rise in peripheral venous pressure.

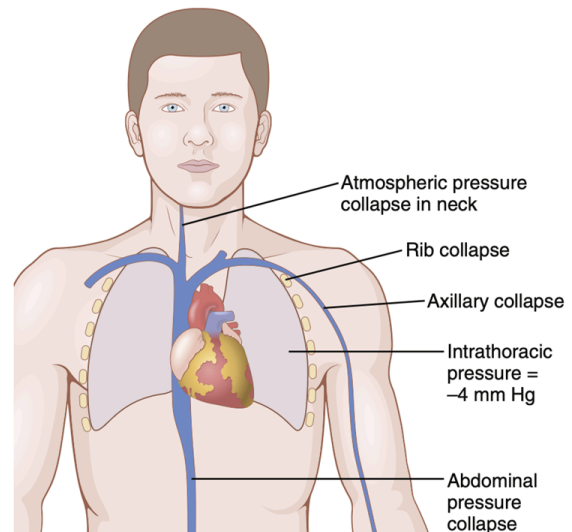


Figure 15-9. Compression points that tend to collapse the veins entering the thorax.

Intro-abdominal Pressure

- Normal: +6 mmHg (recumbent).
- Can rise to +15–30 mmHg in pregnancy, obesity, ascites, tumors.

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- **Femoral venous pressure must exceed intra-abdominal pressure** for blood to flow toward the heart.

Gravitational/Hydrostatic Pressure

- Blood pressure increases **1 mmHg per 13.6 mm vertical distance** due to gravity.
- Standing person:
 - RAP remains ~0 mmHg (heart pumps excess).
 - Foot veins: ~+90 mmHg due to hydrostatic column.
 - Hand veins: ~+35 mmHg (rib compression + hydrostatic effect).
- Neck veins collapse at 0 mmHg (prevent pressure rise/fall).
- **Dural sinuses (inside skull):** can reach -10 mmHg (risk of air embolism if opened).

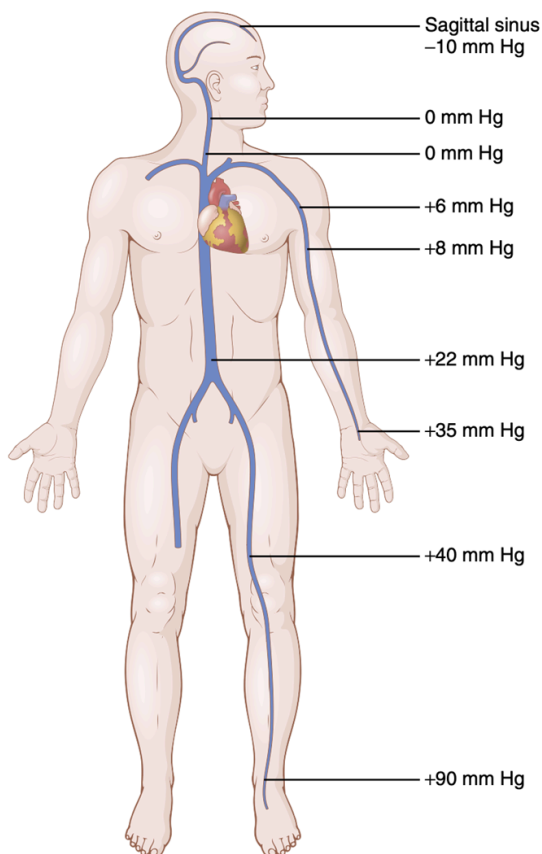


Figure 15-10. Effect of gravitational pressure on the venous pressures throughout the body in the standing person.

Effect on Arterial pressures

- Gravity also affects arterial and capillary pressures.
- Example: MAP = 100 mmHg at heart → arterial pressure at feet ≈ 190 mmHg.
- Reported arterial pressure values are relative to heart level.

VENOUS VALVES & VENOUS PUMP

- **Valves prevent backflow** → ensure one-way flow toward the heart.
- **Muscle contractions compress veins** → blood is propelled forward (muscle/venous pump).
- Walking/moving **keeps venous pressure in feet < 20 mmHg**.
- Standing still: venous pressure in feet rises to +90 mmHg → ↑ capillary pressure → fluid leakage → leg swelling → ↓ blood volume (10–20% lost in 15–30 mins) → possible fainting.
- Prevented by **flexing leg muscles or bending knees to activate venous pump**.

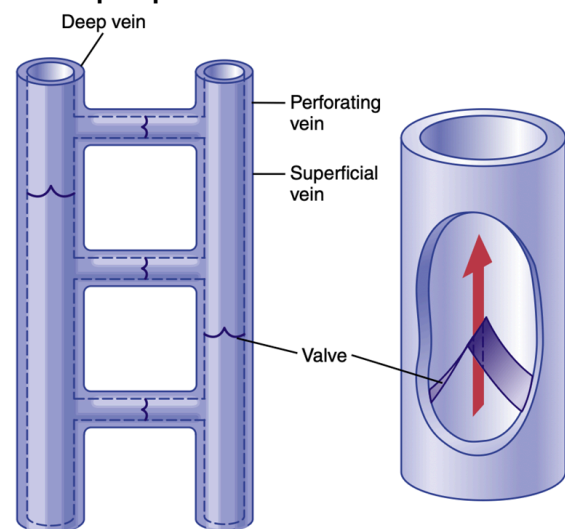


Figure 15-11. Venous valves of the leg.

Venous Valve Incompetence & Varicose Veins

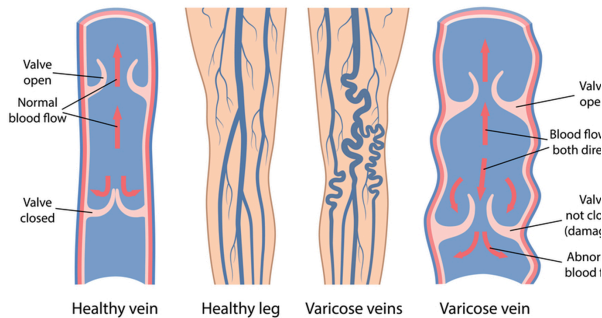
- Chronic high venous pressure (e.g., pregnancy, prolonged standing) → **overstretched veins** → valves fail to close.
- **Leads to varicose veins** (bulging veins, esp. lower legs).
- Consequences:
 - High venous & capillary pressures when standing.
 - Chronic edema → impaired nutrient diffusion.

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- Muscle pain/weakness, skin ulceration, gangrene.
- Management: leg elevation, compression stockings/binders.



Blood Reservoir Function of Veins

- Veins contain **~60% of total blood volume**.
- High compliance allows them to act as a blood reservoir.
- Blood loss → ↓ arterial pressure → **baroreceptors** (carotid sinuses, etc.) → sympathetic activation → venoconstriction → mobilizes stored blood.
- Circulation can still function even after ~20% blood loss.

Specific Blood Reservoirs

- **Spleen**: can release ~100 ml blood.
- **Liver sinuses**: several hundred ml.
- **Large abdominal veins**: ~300 ml.
- **Subcutaneous venous plexus**: several hundred ml.
- **Heart**: ~50–100 ml (shrinks with sympathetic activity).
- **Lungs**: ~100–200 ml.

Spleen as a Reservoir for RBCs

- Venous sinuses: store whole blood.
- **Red pulp**: stores concentrated RBCs trapped in **trabeculae**.
 - Sympathetic activation → contraction → releases ~50 ml RBCs, raising hematocrit by 1–2%.
- **White pulp**: contains lymphoid tissue → part of the immune system (produces lymphocytes).

Key Takeaways:

- Veins act as reservoirs and **regulators of cardiac output**.

- Right atrial pressure (CVP) is the **central reference point for venous pressures**.
- Gravitational and intra-abdominal forces strongly **influence venous return**.
- Clinical extremes (heart failure, hemorrhage, surgery) show how sensitive venous pressures are.
- Venous valves + muscle pump **prevent excessive pooling and maintain low venous pressures during movement**.
- **Valve failure** → **varicose veins** → chronic edema & tissue damage.
- Veins are the main blood reservoir, **mobilized during blood loss via sympathetic stimulation**.
- Spleen is unique as both a **blood reservoir** and an **RBC reservoir** that can acutely increase hematocrit.

Functions of the Microcirculation

- Primary role: **transport of nutrients to tissues & removal of cellular waste**.
- Local control: **small arterioles regulate blood flow**; tissue conditions (nutrients, metabolites, H^+) directly influence arteriolar diameter.
- Efficiency: ~10 billion capillaries with a surface area of 500–700 m^2 → ensures nearly every cell is within 20–30 μm of a capillary.

Organization of Microcirculation

- Arterioles:
 - Highly muscular, diameter = 10–15 μm .
 - Branch into smaller arterioles (5–9 μm).
- **Metarterioles (terminal arterioles)**: intermittent smooth muscle fibers instead of continuous coat.
- **Precapillary sphincters**: rings of smooth muscle at capillary origins → open/close to regulate blood flow into capillaries.
- **Venules**: larger than arterioles, weaker muscle coat, but can contract despite lower pressure.
- **Local metabolic control**: metarterioles & sphincters adjust based on tissue metabolic needs.

Structure of the Capillary Wall

- Single layer of endothelial cells + thin basement membrane.
- Wall thickness: ~0.5 μm .

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- Internal diameter: 4–9 μm (just large enough for RBCs to pass).

Capillary Pores & Transport Pathways

- Intercellular clefts:
 - Narrow slits between endothelial cells (6–7 nm wide).
 - Allow diffusion of water, ions, and small solutes.
 - Account for $\sim 1/1000$ of capillary surface area.
- **Plasmalemmal vesicles (caveolae):**
 - Formed from caveolins + cholesterol + sphingolipids.
 - Functions: endocytosis, transcytosis, and possible vesicular channel formation for macromolecule transport.

Specialized Capillary Structures by Organ

- **Brain (Blood-Brain Barrier):**
 - Tight junctions $\rightarrow$ only very small molecules (H_2O , O_2 , CO_2) pass.
- **Liver:**
 - Large open clefts $\rightarrow$ nearly all dissolved substances, including plasma proteins, can pass.
- **Gastrointestinal tract:**
 - Intermediate pore size between muscle and liver.
- **Kidney glomeruli:**
 - Numerous fenestrae (small oval windows) in endothelial cells.
 - Allow large amounts of small molecules/ions to filter, but exclude plasma proteins.

Flow of Blood in Capillaries - Vasomotion

- Blood flow is intermittent, not continuous, due to vasomotion (cyclical contraction/relaxation of metarterioles & precapillary sphincters).
- **Main regulator:** Oxygen concentration in tissues.
 - $\downarrow$ Oxygen $\rightarrow$ sphincters stay open longer $\rightarrow$ $\uparrow$ blood flow to deliver oxygen & nutrients.
- Although flow in individual capillaries is intermittent, the average function of billions of capillaries results in a stable supply of nutrients and waste removal.

Exchange Between Blood & Interstitial Fluid

- Primary mechanism: Diffusion through capillary membranes.
 - Allows constant exchange of water, electrolytes, nutrients, and waste.
 - Proteins are the major plasma constituents that do not diffuse freely.

Pathways of Diffusion

- **Lipid-soluble substances (O_2 , CO_2):**
 - Diffuse directly through endothelial cell membranes.
 - Very rapid transport due to full membrane permeability.
- **Water-soluble, non-lipid-soluble substances (Na^+ , Cl^- , glucose, water):**
 - Move through intercellular clefts/pores.
 - Despite pores being only $1/1000$ of surface area, rapid thermal motion allows massive exchange.
 - Example: Water exchanges ~ 80 times faster than plasma flows linearly along a capillary.

Effect of Molecular Size

- Capillary pore width: $\sim 6\text{--}7$ nm.
- Size matters:
 - **Small molecules** (water, ions) $\rightarrow$ pass easily.
 - **Medium molecules** (glucose, urea) $\rightarrow$ moderate passage.
 - **Large molecules** (proteins, especially albumin) $\rightarrow$ very limited passage.

Table 16-1 Relative Permeability of Skeletal Muscle Capillary Pores to Different-Sized Molecules

Substance	Molecular Weight	Permeability
Water	18	1.00
NaCl	58.5	0.96
Urea	60	0.8
Glucose	180	0.6
Sucrose	342	0.4
Inulin	5000	0.2
Myoglobin	17,600	0.03
Hemoglobin	68,000	0.01
Albumin	69,000	0.001

Organ Specific Permeability

- Brain: Tight junctions $\rightarrow$ only very small molecules (H_2O , O_2 , CO_2).

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- **Liver:** Very permeable → even plasma proteins pass freely.
- **Kidney (glomeruli):** Extremely permeable to water/electrolytes ($\approx 500\times$ muscle capillaries) but not to proteins.
- **GI tract:** Intermediate permeability.

Concentration Gradient Principle

- **Diffusion rate** $\propto$ **concentration difference across the membrane.**
 - **O₂:** High in blood → diffuses into tissues.
 - **CO₂:** High in tissues → diffuses into blood.
- Even small gradients are enough to supply tissues with required oxygen and remove CO₂ efficiently.

Interstitialium & Interstitial Fluid

- **Interstitialium** = space between cells ($\approx 1/6$ of body volume).
- **Interstitial fluid** = fluid in this space, derived from capillary filtration/diffusion, similar to plasma but with lower protein content.
- **Structural Components**
 - **Collagen fiber bundles** – provide tensile strength.
 - **Proteoglycan filaments** (98% hyaluronic acid, 2% protein) – form a fine reticular network (“brush pile”).
- **Tissue Gel**
 - Interstitial fluid is entrapped in the proteoglycan network, forming a gel.
 - **Fluid movement mainly occurs by diffusion** (not bulk flow).
 - **Diffusion efficiency:** 95–99% as rapid as free fluid → allows fast exchange of water, electrolytes, nutrients, O₂, CO₂, and wastes.
- **Free Fluid**
 - Normally very little (<1% of interstitial fluid).
 - Flows in small rivulets along collagen or cell surfaces.
 - In edema, **free fluid pockets expand greatly**, leading to fluid accumulation independent of the proteoglycan matrix.

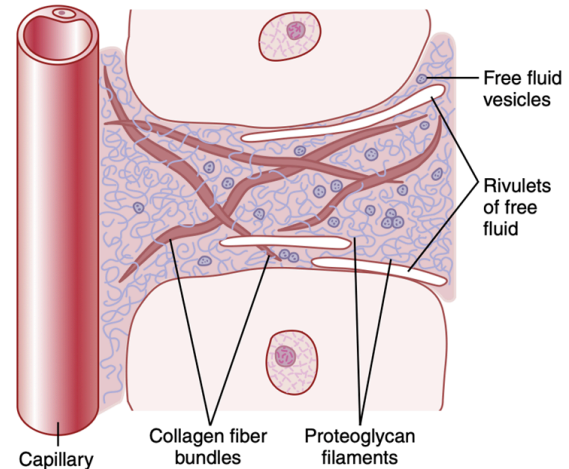


Figure 16-4. Structure of the interstitium. Proteoglycan filaments are everywhere in the spaces between the collagen fiber bundles. Free fluid vesicles and small amounts of free fluid in the form of rivulets occasionally also occur.

Fluid Filtration Across Capillaries

- Determined by **hydrostatic pressure vs. colloid osmotic pressure** → the balance described by **Starling Forces**.
- **Capillary hydrostatic pressure (P_c):** pushes fluid out of capillary.
- **Interstitial hydrostatic pressure (P_{if}):**
 - Positive → pushes fluid into capillary.
 - Negative → pulls fluid out.
- **Plasma colloid osmotic pressure (π_p):** pulls fluid into capillary (mainly due to plasma proteins).
- **Interstitial colloid osmotic pressure (π_{if}):** pulls fluid out into interstitium.
- **Net Filtration Pressure (NFP):**
 - $NFP = P_c - P_{if} - \pi_p + \pi_{if}$
- **Positive NFP** → filtration into interstitium.
- **Negative NFP** → absorption into capillaries.
- **Capillary Filtration Coefficient (K_f):** reflects permeability & surface area of capillaries.
 - $Filtration = K_f \times NFP$

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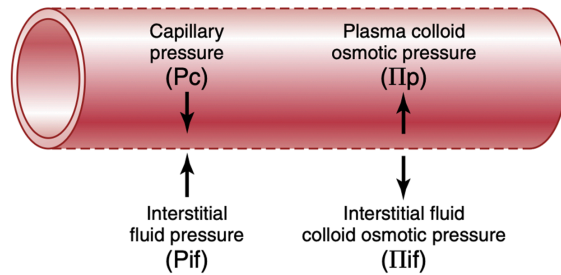


Figure 16-5. Fluid pressure and colloid osmotic pressure forces operate at the capillary membrane and tend to move fluid outward or inward through the membrane pores.

Capillary & Interstitial Pressures

- **Capillary hydrostatic pressure (Pc):** $\approx 17\text{--}25$ mmHg (varies by tissue).
- **Interstitial fluid pressure (Pif):** usually slightly negative (~ -3 mmHg in subcutaneous tissue).
 - Positive in encapsulated organs (kidney, brain, eye, muscles).
- **Examples of negative pressures:**
 - Intrapleural space: -8 mmHg
 - Joint synovial space: -4 to -6 mmHg
 - Epidural space: -4 to -6 mmHg
- **Role of Lymphatic System**
- Acts as a scavenger system $\rightarrow$ removes excess fluid, proteins, debris.
- Pumping of lymphatics helps maintain the slight negative interstitial pressure.

Colloid Osmotic Pressure (Oncotic Pressure)

- Plasma proteins (mainly albumin) create osmotic pressure.
- Normal plasma colloid osmotic pressure: ~ 28 mmHg
 - 19 mmHg from proteins themselves.
 - 9 mmHg from Donnan effect (extra osmotic effect of ions bound to proteins).
- **Interstitial Colloid Osmotic Pressure**
 - Small amounts of protein leak into interstitium via pores & vesicles.
 - Average interstitial protein concentration: ~ 3 g/dl ($\sim 40\%$ of plasma).
 - Corresponding colloid osmotic pressure: ~ 8 mmHg.

Starling Forces Govern Fluid Exchange

- Fluid exchange depends on the balance of hydrostatic pressures and colloid osmotic pressures across the capillary membrane.
- Movement is outward at the arterial end (filtration) and inward at the venous end (reabsorption).

At the Arterial End (Filtration)

- **Outward forces:**
 - Capillary hydrostatic pressure = **30 mmHg**
 - Negative interstitial hydrostatic pressure = **3 mmHg**
 - Interstitial colloid osmotic pressure = **8 mmHg**
 - Total outward = **41 mmHg**
- **Inward force:**
 - Plasma colloid osmotic pressure = **28 mmHg**
 - Total inward = **28 mmHg**
- Net filtration pressure = **+13 mmHg** $\rightarrow$ fluid moves OUT
 - About $1/200$ of plasma volume filters into interstitial fluid each pass.

At the Venous End (Reabsorption)

- **Inward force:**
 - Plasma colloid osmotic pressure = **28 mmHg**
 - Total inward = **28 mmHg**
- **Outward forces:**
 - Capillary hydrostatic pressure = **10 mmHg**
 - Negative interstitial hydrostatic pressure = **3 mmHg**
 - Interstitial colloid osmotic pressure = **8 mmHg**
 - Total outward = **21 mmHg**
- Net reabsorption pressure = **+7 mmHg** $\rightarrow$ fluid moves IN

Overall Fluid Balance

- **$\sim 90\%$ of filtered fluid at the arterial end is reabsorbed at the venous end.**
- **$\sim 10\%$ enters the lymphatic system,** which returns it to circulation.
- This balance maintains normal fluid distribution between plasma and interstitial fluid.
- Essential to Remember:
 - **Filtration (arterial end) > Reabsorption (venous end)**

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- Lymphatic system is critical to return excess fluid and prevent edema.

Starling Equilibrium Concept

- **Ernest Starling's principle:** Capillaries are normally in a state of near-equilibrium.
- Filtration $\approx$ Reabsorption $\rightarrow$ Slight imbalance exists.
- This slight excess filtration is drained by the lymphatic system.

Forces in Starling Equilibrium (Mean Values)

- **Outward forces (28.3 mmHg total):**
 - Capillary hydrostatic pressure = 17.3 mmHg
 - Negative interstitial fluid hydrostatic pressure = 3.0 mmHg
 - Interstitial fluid colloid osmotic pressure = 8.0 mmHg
- **Inward force (28.0 mmHg total):**
 - Plasma colloid osmotic pressure = 28.0 mmHg
- Net Force = +0.3 mmHg $\rightarrow$ Slight net outward filtration

Net Filtration and Lymphatic Role

- Net filtration pressure (0.3 mmHg) $\rightarrow$ About 2 ml/min of fluid (excluding kidneys).
- This excess is normally returned via the lymphatic system.

Capillary Filtration Coefficient (Kf)

- Defines capacity of capillaries to filter fluid:
 - Whole body Kf = 6.67 ml/min per mmHg imbalance.
 - Varies greatly by tissue:
 - **Very small** $\rightarrow$ Brain, muscle
 - **Moderate** $\rightarrow$ Subcutaneous tissue
 - **Large** $\rightarrow$ Intestine
 - **Extremely large** $\rightarrow$ Liver, kidney glomeruli
- Protein concentrations in interstitial fluid also vary:
 - Muscle = 1.5 g/dl
 - Subcutaneous tissue = 2 g/dl
 - Intestine = 4 g/dl
 - Liver = 6 g/dl

Effects of Abnormal imbalances

- $\uparrow$ Capillary pressure $\rightarrow$ Increased filtration $\rightarrow$ **If lymphatics cannot compensate $\rightarrow$ Edema.**
 - Example: +20 mmHg rise $\rightarrow$ Net filtration pressure increases to 20.3 mmHg ($\approx$ 68x normal).
- $\downarrow$ Capillary pressure $\rightarrow$ Net reabsorption $\rightarrow$ **Blood volume increases at expense of interstitial fluid.**

Key Takeaways:

- Microcirculation = nutrient delivery + waste removal, tightly matched to tissue needs.
- Arterioles, metarterioles, and precapillary sphincters are the main regulators of local blood flow.
- Capillary walls are ultra-thin & highly permeable, with organ-specific adaptations (e.g., BBB vs. liver sinusoids vs. kidney fenestrae).
- **Vasomotion** ensures oxygen delivery matches tissue demand.
- **Diffusion** is the dominant exchange mechanism, regulated by solubility, molecular size, and organ-specific permeability.
- Concentration gradients drive efficient bidirectional exchange (O_2 to tissues, CO_2 to blood).
- **Tissue gel structure** enables efficient diffusion-based exchange.
- **Starling forces** determine whether fluid moves out of capillaries (filtration) or back in (absorption).
- Plasma proteins are the major force keeping fluid within circulation.
- **Lymphatic system** prevents edema by removing excess fluid/protein and maintaining negative interstitial pressure.
- **Starling equilibrium** ensures stable fluid balance between plasma $\leftrightarrow$ interstitial fluid, with lymphatics correcting the slight excess filtration. Disruption of this balance (too much filtration or reabsorption) leads to edema or altered blood volume.

Regulation of Blood Pressure

- **Vasomotor Area of the Medulla**
 - Center responsible for regulation of HR and BP
 - Found in the Medulla
 - Lateral portion: Excitatory area (increase in HR and BP)

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- **Medial Inhibitory Area** (decrease HR and BP)
 - Controlled by the Hypothalamus and other higher nervous centers
- **Acute Control**
 - ANS control, Baroreceptors, Chemoreceptors, Low-pressure receptors, CNS Ischemic response.

Regulation of Blood Pressure

- **Long-term control**
 - Renin-Angiotensin-Aldosterone System (RAAS)
- **ANS**
 - Sympathetic > Parasympathetic
 - To increase BP via ANS:
 - **Arteriolar Vasoconstriction** increase TPR increase BP
 - **Venous Vasoconstriction** increase VR increase CO increase BP
 - **Increase HR and SV via B1 receptors of the Heart** increase CO increase BP
- **Baroreceptors**
 - Act fast; buffers minute-to-minute changes in BP
 - Stretch receptors on the Carotid Sinus and Aortic Arch
 - inc. BP inc. Stretch inc. firing of CN IX to Nucleus Tractus Solitarius trigger parasympathetic response
 - **Carotid baroreceptors:** respond to increase/decreases in pressure from 50mmHg – 180mmHg
 - **Aortic baroreceptors:** respond to increase in pressure >80mmHg
 - **Set point for Mean Arterial Pressure in Vasomotor Center:** 100mmHg

- **Valsalva Maneuver**
 - Expiring against closed Glottis inc. intrathoracic pressure decrease Venous Return (VR) dec. CO and BP sensed by Baroreceptors inc. HR
- **Chemoreceptors**
 - Could also induce changes in BP
 - Respond to low Oxygen, high Carbon dioxide concentration whenever BP <80mmHg
 - When chemoreceptors are stimulated, they activate vasomotor centers for systemic vasoconstriction inc. TPR inc. BP
- **Low-pressure Receptors (Cardiopulmonary Baroreceptors)**
 - In the atria and pulmonary arteries
 - Detects “fullness” of vascular system (increase intravascular volume)
 - In response to increased intravascular volume:
 - Increase Atrial Natriuretic Peptide: inc. Sodium and water excretion
 - **In dehydration,** there is increased ADH, Angiotensin II, Aldosterone, Norepinephrine, and decrease ANP
 - ANP is released in response to increased atrial pressure from increased blood volume
 - **Decrease Anti-Diuretic Hormone (ADH):** inc. urine output
 - **Renal vasodilation:** increase urine output
 - **Increase HR:** helps match VR with CO.

INTEGRATIVE PHYSIOLOGY (POST LEC)

Lecture #2: CIRCULATORY & MUSCLE | OPHT 1-1 | MIDTERMS

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- **CNS Ischemic Response**
 - The vasomotor center itself responds directly to the ischemia during low BP
 - Starts at BP<60mmHg and optimal at a BP=15-20mmHg
 - **The “last-ditch” stand:**
 - All systemic arterioles vasoconstrict severely EXCEPT for Coronary vessels, and Cerebral vessels
- **Long-term control**
 - Renin-Angiotensin-Aldosterone System (RAAS)
 - Slow: takes 20 minutes to take effect; takes several hours for optimal effects
 - Activated when faster mechanisms (e.g., baroreceptors) fail to regulate BP
 - Also responsible for maintaining normal BP despite wide variation in salt intake

Renin-Angiotensin-Aldosterone System (RAAS)

- Activated first before RAAS: **Baroreceptor reflex**
- Senses decrease in NaCl delivery in Distal Convoluted Tubule (DCT): Macula densa
- **Stimulates Juxtaglomerular (JG) Cells to release Renin**
 - Also increases renin secretion: Sympathetic nerve activity by Beta 1 receptors)
- Converts Liver Angiotensinogen to Angiotensin 1: Renin
- Converts Angiotensin I to Angiotensin II: Lung Angiotensin-Converting Enzyme
- Converts Angiotensin II effects
- Maintains (preserves) normal Glomerular Filtration Rate (vasoconstricts efferent > afferent)
- Vasoconstricts systemic arterioles increase TPR
- Stimulates Aldosterone secretion by Adrenal cortex
- Increase Na-H exchange in the Proximal Convoluted Tubule contraction alkalosis
- Increase thirst and ADH secretion increase intravascular volume

- Increase Epinephrine, Norepinephrine, Cortisol secretion vasoconstricts arterioles
- Aldosterone effects on the Kidneys:
 - Sodium and water reabsorption
 - Potassium secretion
 - Hydrogen secretion

HORMONAL / HUMORAL MECHANISMS FOR BLOOD FLOW CONTROL

VASOCONSTRICTORS	VASODILATORS
• Vasopressin: most potent vasoconstrictor	• Prostacyclin (PGI₂): counteracts TXA₂
• Serotonin: released because of blood vessel damage; causes arteriolar vasoconstriction; implicated in migraine	• Nitric Oxide (NO): vasodilates upstream blood vessels ○ MOA: guanylate cyclase and cGMP ○ Acetylcholine causes vasodilation by increasing production of NO in vascular smooth muscle ○ Acetylcholine causes VASOCONSTRICTION instead of vasodilation whenever the endothelium is damaged due to decreased NO
• Endothelin: released by damaged endothelium	• PGE: vasodilators
• PGF and Thromboxane A₂ (TXA₂)	• Lactate, Adenosine found in muscles
• Norepinephrine, Epinephrine	• Bradykinin and Histamine causes arteriolar dilation & venous constriction leading to increased filtration (local edema)
• Angiotensin II	• H⁺, CO₂, K, ANP

INTEGRATIVE PHYSIOLOGY (POST LEC)

Lecture #2: CIRCULATORY & MUSCLE | OPHT 1-1 | MIDTERMS

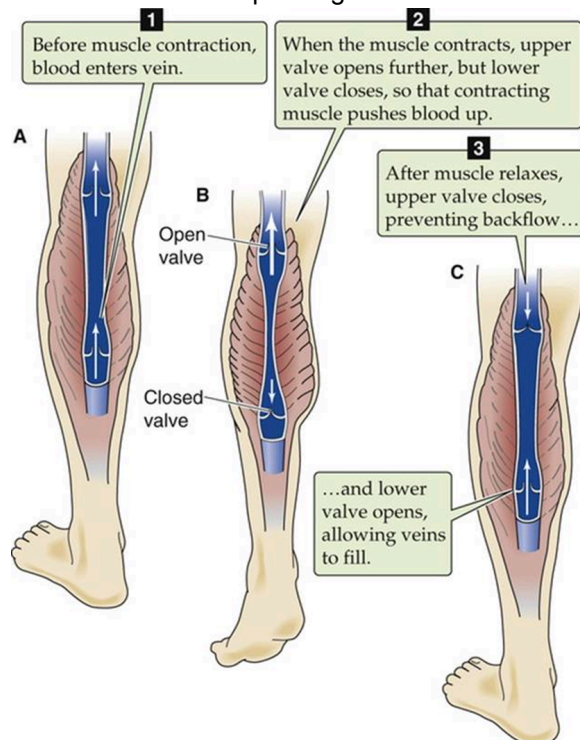
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MUSCLE BLOOD FLOW

- **At rest:**
 - Blood flow $\approx$ 3–4 mL/min/100 g muscle
- **During extreme exercise (well-trained athletes):**
 - Blood flow can $\uparrow$ 25–50 fold $\rightarrow$ 100–200 mL/min/100 g muscle
 - Peak values up to 400 mL/min/100 g (thigh muscles)

Blood Flow Changes During Muscle Contraction

- **Rhythmic contraction:**
 - Blood flow fluctuates $\rightarrow$ $\downarrow$ during contraction (vessel compression), $\uparrow$ after contraction.
- **Tetanic contraction:**
 - Sustained compression $\rightarrow$ flow can be almost stopped, leading to rapid fatigue.



Capillary Recruitment in Exercise

- **At rest:** Many muscle capillaries are dormant.
- **During exercise:** Nearly all capillaries open, which:

- $\downarrow$ diffusion distance for O_2 & nutrients
- $\uparrow$ capillary surface area 2–3 fold for exchange

Local (Metabolic) Control of Blood Flow

- **Primary mechanism:** $\downarrow$ oxygen levels $\rightarrow$ arteriolar vasodilation
- **Vasodilator factors released during activity:**
 - Adenosine (important but limited effect)
 - K^+ ions
 - ATP
 - Lactic acid
 - CO_2
- These substances sustain increased blood flow as long as exercise continues.

Nervous Control of Muscle Blood Flow

- **Sympathetic vasoconstrictor fibers:**
 - Release norepinephrine $\rightarrow$ $\downarrow$ blood flow to $\frac{1}{2}$ – $\frac{1}{3}$ normal in resting muscle.
 - Helps maintain arterial pressure in shock/stress.
- **Adrenal medulla secretion during exercise:**
 - **Norepinephrine** $\rightarrow$ vasoconstriction (via α -receptors)
 - **Epinephrine** $\rightarrow$ slight vasodilation (via β -receptors)

Three Major Circulatory Adjustments in Exercise

- **Sympathetic nervous system activation** $\rightarrow$ strong stimulatory effects.
- **Increase in arterial pressure** $\rightarrow$ ensures adequate perfusion.
- **Increase in cardiac output** $\rightarrow$ supports high muscle demand.

Effects of Sympathetic Activation

- **Heart effects:**
 - $\uparrow$ Heart rate and contractility due to sympathetic drive.
 - $\downarrow$ Parasympathetic inhibition $\rightarrow$ stronger cardiac output.
- **Arteriolar changes:**
 - Vasoconstriction in most peripheral tissues.
 - Vasodilation in active muscles (local vasodilators dominate).

INTEGRATIVE PHYSIOLOGY (POST LEC)

Lecture #2: CIRCULATORY & MUSCLE | OPHT 1-1 | MIDTERMS

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- Blood is “redistributed” → less to nonessential areas, more to muscles (+ up to 2 L/min extra flow).
- Coronary and cerebral circulation spared → protected from vasoconstriction.

Effects of Sympathetic Activation

- **Venous changes:**
 - **Venoconstriction** → ↑ mean systemic filling pressure.
 - ↑ **Venous return** → boosts cardiac output.

Sympathetic Stimulation & Arterial Pressure

- Mechanisms for ↑ arterial pressure:
 - **Vasoconstriction in most tissues** (except brain, heart, and active muscles).
 - Stronger pumping activity of the heart.
 - ↑ Venous return due to venous contraction.
- **Range of arterial pressure increase:**
 - +20 to +80 mmHg depending on exercise conditions.
- **Specific scenarios:**
 - Tense exercise with few muscles active → vasoconstriction dominates → MAP can rise to ~170 mmHg.
 - Whole-body exercise (running, swimming) → widespread vasodilation → pressure rise only 20–40 mmHg.
- Essential concept:
- During exercise, **the body balances widespread sympathetic activation (raising HR, BP, venous return) with local vasodilator effects in active muscles**, ensuring blood is redirected to where it is most needed without compromising the heart and brain.

Parameter	At Rest	During Exercise	Key Concept
Heart rate (HR)	Normal (60–80 bpm)	↑ significantly	Sympathetic activation + ↓ parasympathetic tone
Cardiac output (CO)	~5 L/min	↑ 4–7x (up to 25–30 L/min in athletes)	Increased HR + stroke volume
Stroke volume (SV)	Normal	↑ initially (then plateaus at high HR)	Stronger contraction + ↑ venous return
Arterial pressure (MAP)	~90–100 mmHg	↑ 20–80 mmHg	Sympathetic vasoconstriction + CO rise
Arterioles	Balanced tone	Vasoconstriction in nonactive tissues; vasodilation in active muscles	Redistribution of blood
Blood flow to muscles	15–20% of CO	80–85% of CO	Local vasodilators (+O ₂ , ↑CO ₂ , ↑K ⁺ , lactic acid, adenosine)
Blood flow to brain	Constant (~750 ml/min)	Constant	Cerebral perfusion preserved
Blood flow to heart (coronary)	~250 ml/min	↑ 3–4x	Matches ↑ myocardial O ₂ demand
Venous return	Baseline	↑ greatly	Venoconstriction + muscle pump
Capillary recruitment in muscle	Limited	Almost all open	Shorter O ₂ diffusion distance, ↑ surface area

Why Increased Arterial pressure During Exercise is Important

- Muscle blood flow during exercise depends on arterial pressure rise.
- Example: A 30% rise in arterial pressure → 30% more driving force for blood flow.
- Increased arterial pressure stretches vessel walls, enhancing flow with local vasodilators.
- Marathon runners: blood flow can ↑ 20 times normal due to combined effects.
- Cardiac output (CO) must rise proportionally to exercise intensity.
- The ability of circulation to deliver oxygen and nutrients is as important as muscle strength.
- High cardiac output capacity = better endurance and athletic performance.
- Marathon runners with the highest CO often set record-breaking times.

Graphic Analysis of Cardiac Output Changes

- Cardiac output and venous return curves intersect at equilibrium points:
 - Point A = rest
 - Point B = heavy exercise
- Increased CO requires changes in venous return → cannot occur without adjustments.

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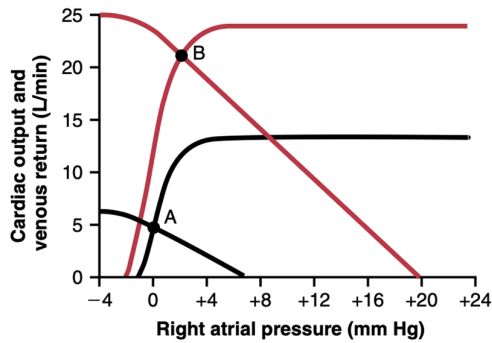


Figure 21-2. Graphic analysis of change in cardiac output, venous return, and right atrial pressure with the onset of strenuous exercise. Black curves, Normal circulation. Red curves, heavy exercise.

Two Key Adjustments During Heavy Exercise

- **↑ Mean Systemic Filling Pressure (MSFP):**
 - Caused by sympathetic venoconstriction + abdominal and skeletal muscle compression.
 - MSFP can rise from 7 mmHg → 30 mmHg, enhancing venous return.
- **↑ Slope of Venous Return Curve:**
 - Caused by decreased resistance in active muscle vessels.
 - ↓ Resistance → easier venous return → steeper slope of venous return curve.

Venous Return & Cardiac Output Curves – New Equilibrium (Exercise vs Rest)

- Equilibrium shift:
 - Rest → Point A
 - Heavy exercise → Point B (↑ cardiac output, minimal change in right atrial pressure).
- **Right atrial pressure (RAP):**
 - In healthy individuals → RAP rises only ~1.5 mmHg during heavy exercise.
 - In strong hearts → RAP may even fall below normal due to strong sympathetic stimulation.
 - In weak hearts → even moderate exercise can cause a marked rise in RAP.

Key Takeaways:

- **Exercise blood flow demand is massive**
→ Cardiac output must ↑ up to 6–7× normal in trained athletes.
- **Metabolic control** (low O_2 + vasodilators) is the main driver of increased flow.
- **Sympathetic control fine-tunes distribution:** vasoconstriction preserves blood pressure, while epinephrine provides mild vasodilation.
- **Capillary recruitment ensures efficient O_2 /nutrient delivery** during activity.
- Combination of ↑ MSFP + ↓ venous resistance → greatly enhances venous return.
- This supports increased cardiac output, allowing oxygen delivery to match exercise demands.
- During exercise, the **circulatory system can greatly increase cardiac output with minimal increase in right atrial pressure** if the heart is strong.
- **Right atrial pressure response** is a marker of cardiac strength:
 - **Low/unchanged RAP** → strong heart
 - **High RAP** → weakened heart.